GREEN TECHNOLOGY

SOLAR POWERED INDIA

"I'd put my money on the sun and solar energy. What a source of power! I hope we don't have to wait until oil and coal run out before we tackle that."– Thomas Edison

India is a country with abundant natural resources, and solar power is not left behind. In a recent document published by the Ministry of New and Renewable Energy (MNRE), it has been established that India today receives more than 5,000 trillion kWh/year of solar radiation, which is far more than its total annual energy requirement. The radiation can have various usages in thermal and voltaic application areas.

Solar thermal technology uses the sun's energy, rather than fossil fuels, to generate low-cost environment-friendly thermal energy. This energy is used to heat water or other fluids for industrial, commercial and domestic sectors of the country. In addition, it can power solar cooling systems. In new-age civil construction setups, solar-powered steam-generating systems and air-heating technologies are in vogue, especially in urban landscapes.

Solar photovoltaic technologies have their own application areas and these are used majorly for solar lanterns, solar home systems, solar streetlights, solar pumps, solar power packs, rooftop SPV systems, etc, reducing the burden on conventional fuels. So what's the future ahead?

Wider acceptance of solar thermal systems drives business value in the long run by providing the following set of benefits:

- **Reduced bills:** Large business houses that require many gallons of hot water and other fluids must pay higher amounts for fuel-heating requirements, whereas manufacturing units using solar power save nearly 70 per cent on this cost.
- **Benefits of compliance with government mandates:** Many areas and green zones must meet some government mandates and solar thermal systems help meet the requirements set. This also benefits them in terms of sops from the government while also giving a ROI.
- **Reduction in carbon footprint:** By utilising solar energy instead of fossil fuels, solar thermal systems reduce the amount of onsite-generated, carbon-based greenhouse gases that businesses emit into the atmosphere.

Early 2010, the government of India, realising the benefits of solar power, launched the National Solar Mission with an initial target of deploying 20,000MW of grid-connected solar power by 2022. The idea was to reduce the cost of solar power generation through the following measures:

- Long-term policy framework to